

Justice as a Governance Capability: A Computational Systems Model of Multi-Level Governance for Human Flourishing under Bounded Non-Exploitation

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Abstract

Most formal models of governance treat governance as a single-level phenomenon and permit jurisdictions to advance their own flourishing at the expense of others. This paper extends the General Governance Success Model (GSM) to multi-level governance under a bounded non-exploitation constraint: no jurisdiction may impose uncompensated, non-trivial sacrifices on another for its own benefit. The extension introduces justice as a sixth governance capability, decomposed into distributive, procedural, recognition, corrective, and intergenerational dimensions. Justice is treated as a productive, fragile, multi-level capability that co-evolves with accountability, competence, cohesion, continuity, and learning—not merely as an external normative constraint. Four constraints govern the justice subsystem: bounded non-exploitation, a dimension-specific justice floor, vulnerability-sensitive non-exploitation, and vertical alignment enforced dimension-wise. A signed justice interaction matrix captures synergies and tensions between dimensions; the matrix is disciplined by bounded support, sign priors, and a hierarchical Bayesian identification protocol. Cross-level accountability is specified at the dimension level, so that recognition and intergenerational shortfalls cannot be masked by high distributive performance. Constraint enforcement is modelled as policy intervention with designated authority, implementation cost distributed across multiple capability channels, finite horizon, and passive recovery, yielding an endogenous frequency ceiling on aggressive justice improvement. A dimension-specific prioritarian welfare function completes the architecture. GSM-J is a conditional normative systems model: it makes value commitments explicit and swappable, but does not derive them. Its contribution is a formal substrate for a computational theory of normative governance systems. The model is released as an open-source Python library, `gsm-governance`.

Keywords: multi-level governance; justice as capability; bounded non-exploitation; computational normative governance; polycentric governance; prioritarian welfare; recognition justice; intergenerational justice; complex adaptive systems

1. Introduction

1.1 The Challenge of Multi-Level Governance

Contemporary governance problems—climate change, pandemic response, migration, digital regulation, financial stability—are inherently multi-level. Households, communities, municipalities, provinces, nation-states, regional blocs, and global institutions act in coupled ways. Yet the dominant formal models in political economy and public choice treat governance as occurring at a single level, typically the nation-state.

The multi-level structure matters because it creates opportunities for **jurisdictional exploitation**: a higher level imposing uncompensated costs on a lower level, a powerful state extracting from a weaker one, a present generation depleting the capabilities of a future one. These are governance failures arising from asymmetric power, incomplete representation, and the absence of enforceable reciprocity. Most formal models either assume them away or permit them when aggregate welfare improves—endorsing arrangements that are efficient but unjust.

1.2 The Computational and Systems Perspective

This paper adopts a computational systems approach. Governance is modelled as a complex adaptive system with multiple interacting levels, feedback loops, and emergent dynamics. It builds on six traditions:

1. **Systems theory and cybernetics** (Ashby, 1956; Beer, 1972; Meadows, 2008).
2. **Polycentric governance** (Ostrom, 1990, 2010; McGinnis, 1999).
3. **Institutional economics** (North, 1990; Acemoglu & Robinson, 2012).
4. **Capability theory** (Sen, 1999; Nussbaum, 2011).
5. **Eudaimonic well-being research** (Ryan & Deci, 2001; Seligman, 2011).
6. **Contemporary justice theory** (Rawls, 1971; Fraser, 2009; Young, 1990; Pettit, 1997; Gardiner, 2011).

1.3 The Justice Gap

Bounded non-exploitation is not coextensive with justice. A system can satisfy it perfectly while remaining unjust in at least five respects:

1. **Distributive injustice below the exploitation threshold.**
2. **Procedural exclusion without extraction.**
3. **Recognition injustice without resource harm.**
4. **Corrective neglect** of ended but unremedied harm.
5. **Intergenerational depletion** through voluntary transactions.

These are the normal form of structural injustice in multi-level systems: harm that operates below the threshold of detectable exploitation, through background institutions and accumulated advantage.

We therefore extend GSM with justice as a sixth capability cluster—productive, fragile, multi-level, and contestable.

1.4 Core Contributions

1. A multi-level governance architecture. GSM-J formalizes *L* interacting levels with coupled jurisdictions, vertical and horizontal interaction decomposed into cooperative and exploitative components, and cross-level information flows with fidelity matrices.

2. Justice as a governance capability. Justice is decomposed into five dimensions with a signed interaction matrix, embedded in the state space, and coupled to the other five capabilities. This shifts justice from an evaluative afterthought to an endogenous state variable.

3. An enforcement and accountability architecture. Four justice constraints—bounded non-exploitation, dimension-specific floor, vulnerability-sensitive non-exploitation, and dimension-wise vertical alignment—are complemented by a dimension-specific accountability mechanism and a policy-intervention projection operator with designated authority, multi-channel implementation cost, finite horizon, and passive recovery.

4. A conditional normative framework. Normative parameters (weights, floors, prioritarian coefficients, subsidiarity, coupling coefficients, intervention weights) are explicit, representable, and swappable. The model predicts consequences; it does not derive values.

5. An open-source implementation. The `gsm-governance` Python library implements the full model and a scenario protocol that supports ablation, sensitivity analysis, and empirical extension.

1.5 The Normative Status of GSM-J

GSM-J is a conditional model, not a normative theory. Given specified normative parameters, the model predicts systemic consequences. Its contribution is to make those parameters explicit, contestable, and swappable. Users who reject prioritarianism can set $\delta_d = \gamma_d$; users who reject intergenerational weighting can set $w_t = 0$; users who reject floors can set $J_{\min}^d = 0$. Each choice produces a different model, and each choice's consequences are inspectable.

The architecture constitutes a candidate framework for a **computational theory of normative governance systems**: a formal apparatus in which normative commitments are explicit parameters, systemic consequences are derivable, and the space of possible governance regimes is explorable by simulation. GSM-J does not resolve normative disagreement; it structures it.

2. Theoretical Foundations

2.1 Human Flourishing

We adopt a eudaimonic, multi-dimensional, capability-based conception of flourishing: the ongoing realization of a good human life, comprising subjective well-being, meaning and purpose, growth and mastery, autonomy and agency, relationships and belonging, health and vitality, character and virtue, engagement and play, and contribution and transcendence. Flourishing is relational, ecological, lifelong, and imperfect.

2.2 Bounded Non-Exploitation

Let $E_{ij}(t) \in \mathbb{R}_{\geq 0}$ denote the exploitation index: the uncompensated, non-trivial sacrifice imposed by jurisdiction i on jurisdiction j at time t , for i 's benefit. The bounded non-exploitation constraint states:

$$E_{ij}(t) \leq \tau_j \text{ or } \text{Comp}_{ij}(t) \geq E_{ij}(t) \quad \forall i, j, t$$

where τ_j is jurisdiction j 's tolerance threshold and Comp_{ij} is the compensation vector transferred from i to j .

2.3 Why Non-Exploitation Is Not Sufficient for Justice

Bounded non-exploitation is negative and dyadic. Justice is broader as shown in table 1.

Table 1. Non-exploitation versus justice: A comparison across five respect dimensions

Respect	Non-exploitation	Justice
Valence	Negative (prohibition)	Positive and negative
Scope	Bilateral dyads	Whole-system distributions
Temporal	Present-focused	Includes intergenerational obligation
Structural	Leaves background institutions unexamined	Interrogates institutional design
Remedial	Compensates harm	Requires correction of historical wrong

A system can satisfy bounded non-exploitation perfectly while remaining profoundly unjust. **Non-exploitation is therefore necessary but not sufficient for justice.**

2.4 Justice as a Governance Capability

Justice is:

- **Productive** — it enables trust, voluntary compliance, information sharing, and long-horizon cooperation.
- **Fragile** — it can be eroded by short-term optimization, power concentration, and exploitation of vulnerable populations.
- **Multi-level** — realized or violated at household through global levels.
- **Contestable** — what counts as just is a matter of normative disagreement.

This conception draws on Rawls (1971), Sen (1999), Nussbaum (2011), Fraser (2009), Young (1990), Pettit (1997), and Gardiner (2011). It is pluralist in recognizing multiple irreducible dimensions and prioritarian in weighting improvements to the worst off more heavily.

3. The Computational Model: GSM-J

3.1 Levels and Units

Let $l \in \{1, \dots, L\}$ index governance levels, $L = 6$, then Six levels are modeled, from household to global commons, each with a representative unit. The multi-level structure is what creates the possibility of jurisdictional exploitation and cross-level justice obligations, as shown table 2.

Table 2. Governance levels in the GSM-J architecture

l	Level	Representative Unit
1	Household / individual	Person, family
2	Community	Neighborhood, village
3	Subnational	Municipality, province
4	National	Nation-state
5	Regional	Supranational bloc
6	Global	Planetary commons

Let $i \in \mathcal{I}^{(l)}$ index jurisdictions at level l ; $\mathcal{N}(i)$ denotes same-level neighbors; $i^{\pm 1}$ denotes parent and child units.

3.2 State Vector

Each jurisdiction i at level l carries:

$$\mathbf{x}_i^{(l)}(t) = [A_i, C_i, S_i, T_i, L_i, J_i]^{(l)}(t) \in [0, 1]^6$$

including accountability (A), institutional competence (C), social cohesion (S), strategic continuity (T), adaptive learning (L), and justice (J). Justice is the newly added sixth cluster; the other five are inherited from the General Governance Success Model, as detailed in table 3.

Table 3. Capability clusters in the state vector

Symbol	Capability Cluster	Description
A_i	Accountability	Answerability of power-holders
C_i	Institutional competence	Design and delivery capacity
S_i	Social cohesion	Trust, shared norms, mutual identification
T_i	Strategic continuity	Consistency of long-horizon commitments
L_i	Adaptive learning	Capacity to update in response to feedback
J_i	Justice	Fairness across five dimensions

3.3 Justice Capability Cluster

$$J_i^{(l)}(t) = [J_i^D, J_i^P, J_i^R, J_i^C, J_i^I]^{(l)}(t)$$

Where each of J^D , J^P , J^R , J^C , and J^I represent one dimension of justice capability, as detailed in table 4.

Table 4. The five dimensions of the justice capability cluster

Symbol	Dimension	Operational Indicator
J^D	Distributive	Capability-adjusted Gini; maximin floor; Palma ratio
J^P	Procedural	Participation; transparency; due-process; contestability
J^R	Recognition	Non-domination, non-exclusion, non-assimilation composite
J^C	Corrective	Reparation coverage; restoration rate
J^I	Intergenerational	Depletion-adjusted savings; carbon/resource debt

Scalar aggregate:

$$J_i^{(l)} = \sum_{d \in \{D, P, R, C, I\}} w_d J_i^{d, (l)}, \quad \sum_d w_d = 1, w_d \geq 0$$

Weights are not derived from the model; users must specify and justify them.

3.3.1 Justice Dimension Interactions: M_J

Justice dimensions interact—not always synergistically. The signed interaction matrix $M_J \in \mathbb{R}^{5 \times 5}$ captures both.

$$c_j^d(J_i) = \sum_{e \neq d} m_{de} J_i^e (1 - J_i^d)$$

where $m_{de} \in [-, m_{\max} + m_{\max}]$ is the signed coefficient from e to d , and $(1 - J_i^d)$ is a saturation factor.

Synergy entries (positive).

Table 5. Synergy entries in the justice interaction matrix M_J

Entry	Theoretical claim
m_{PR}, m_{RP}	Procedural voice and recognition reinforce each other
m_{DP}, m_{PD}	Distributive security and procedural participation reinforce each other
m_{DC}	Distributive capacity enables corrective remedy
m_{CR}, m_{RC}	Corrective remedy and recognition mutually reinforce
m_{IP}	Intergenerational concern extends procedural inclusion

Tension entries (negative).

Table 6. Tension entries in the justice interaction matrix M_J

Entry	Theoretical claim
m_{ID}, m_{DI}	Present-oriented distribution displaces future investment
m_{CD}	Large corrective transfers draw from present distributive resources

Entry	Theoretical claim
m_{PI}, m_{IP}^-	Present participation crowds out future representation

The net coefficient for each ordered pair is the sum of synergy and tension contributions. Default values in gsm-governance are set to theoretical priors; empirically, M_j is estimable.

3.3.2 Estimation Protocol for M_j

Signed coefficients introduce an overparameterization risk. For five dimensions, M_j contains 20 off-diagonal coefficients. Three constraints discipline estimation.

Constraint 1 — Bounded support. $m_{de} \in [-m_{\max}, m_{\max}]$, default $m_{\max} = 0.30$.

Constraint 2 — Sign priors. Each coefficient receives a sign prior $\sigma_{de} \in \{+, -, \pm, 0\}$ from §3.3.1. Priors are informative but not binding.

Constraint 3 — Symmetry regime. Two regimes are available:

- **Symmetric** ($m_{de} = m_{ed}$): 10 free parameters. Appropriate for small samples.
- **Asymmetric** ($m_{de} \neq m_{ed}$): 20 free parameters. Appropriate when directional hypotheses are of interest and data permit.

Estimation strategy. A hierarchical Bayesian dynamic factor model:

1. **Observation model.** Observed indicators $J_{i,t}^d = \text{latent } J_{i,t}^d + \varepsilon_{i,t}^d$, $\varepsilon \sim \mathcal{N}(0, \sigma_d^2)$.
2. **State evolution.** Latent justice follows §4.2 with coupling $\mathcal{C}_j^d$ depending on M_j .
3. **Priors on M_j .** Truncated Gaussians centered on sign-prior means with small scale when the sign is definite, larger when contested.
4. **Partial pooling.** Jurisdiction-specific $m_{de}^{(i)} \sim \mathcal{N}(m_{de}^{\text{group}}, \tau_{de}^2)$.
5. **Identification.** At least one exogenous shock per jurisdiction affecting a subset of dimensions; sufficient temporal variation; sign restrictions to break reflection and permutation symmetries.
6. **Inference.** Particle filter or EKF with importance resampling; MCMC or variational inference for posterior.

Fallback. When users lack panel data, default theoretical priors should be used explicitly, and results reported under at least two alternative priors (all-positive vs. mixed-sign) for sensitivity.

3.3.3 Operationalizing Recognition Justice

$$J^R = \frac{1}{3}(R^{\text{dom}} + R^{\text{exc}} + R^{\text{assim}})$$

Table 7. Components of recognition justice (J^R)

Component	Definition
R^{dom}	Non-domination: absence of arbitrary interference
R^{exc}	Non-exclusion: inclusion in the community of peers
R^{assim}	Non-assimilation: tolerance of difference

Measurement protocols in Appendix C.

3.4 Exploitation Index

$$E_{ij}(t) = \sum_d E_{ij}^d(t)$$

Table 8. Components of the exploitation index (E_{ij})

Component	Measurement
E_{ij}^D	Net resource transfer relative to fair-share benchmark
E_{ij}^P	Asymmetry in decision-making influence
E_{ij}^R	Assimilation pressure; dignity violations
E_{ij}^C	Unremedied harm from prior exploitation
E_{ij}^I	Depletion of future-oriented capital

Non-exploitation in one dimension does not imply non-exploitation in another.

3.5 Compensation Mechanism

$$\text{Comp}_{ij}(t) = [\text{Comp}^F, \text{Comp}^I, \text{Comp}^R, \text{Comp}^{\text{Reg}}, \text{Comp}^{\text{Fut}}, \text{Comp}^{\text{Rest}}]_{ij}(t)$$

Table 9. Components of the compensation vector ($Comp_{ij}$)

Component	Primary Target
Fiscal transfers	J^D
Infrastructural investment	J^D, J^C
Representational inclusion	J^P, J^R
Regulatory constraint	J^P, J^R, J^I
Future-oriented transfer	J^I
Restitutional benefits	J^C

Adequacy: $\lambda_j \cdot Comp_{ij}(t) \geq E_{ij}(t)$, with $\lambda_j \in [0, 1]^6$.

3.6 Justice Constraints

3.6.1 C1 – Bounded Non-Exploitation

$$E_{ij}(t) \leq \tau_j \text{ or } Comp_{ij}(t) \geq E_{ij}(t) \forall i, j, t$$

3.6.2 C2 – Justice Floor

$$J_i^{d,(l)}(t) \geq J_{\min}^d \forall i, l, d, t$$

Philosophical foundation. Three traditions converge on a moral minimum: **sufficientarianism** (Frankfurt, 1987; Crisp, 2003; Shields, 2012), **capability thresholds** (Nussbaum, 2000, 2011), and **human rights frameworks** (UDHR, 1948).

Table 10. Default justice floors ($J_{\min}^d$) by dimension.

Dimension	Default $J_{\min}^d$	Justification
J^D	0.30	Maximin floor at ~30% of capability-adjusted median

Dimension	Default $J_{\min}^d$	Justification
J^P	0.25	Minimal procedural rights
J^R	0.25	Non-domination threshold
J^C	0.10	Minimal restorative commitment
J^I	0.20	Weak sustainability

Defaults are placeholders for deliberation.

3.6.3 C3 — Vulnerability-Sensitive Non-Exploitation

$$\text{If } J_j^{(l)}(t) < J_{\min}^{\text{agg}}, \text{ then } E_{ij}(t) = 0 \forall i$$

This closes a gap in C1: a jurisdiction below the justice floor may have a low tolerance threshold and thus be exploited legally under C1. C3 prohibits this, encoding a prioritarian commitment.

3.6.4 C4 — Vertical Justice Alignment (Subsidiarity)

C4 is a **non-regression principle**, not uniformity.

Dimension-level formulation (default).

$$J_i^{d,(l)}(t) \geq \phi \cdot J_i^{d,(l-1)}(t) \forall d$$

This prevents a parent from satisfying C4 in aggregate while eroding one dimension below the child's level. The aggregate form is available as `C4Mode.AGGREGATE`.

4. Dynamics

4.1 Capability Dynamics

Each capability evolves according to dynamics calibrated to its functional role. Full specifications are in `gsm-governance`.

4.2 Justice Dynamics

$$\frac{dJ_i^d}{dt} = \underbrace{g_d(\mathbf{x}_i)}_{\text{internal}} + \underbrace{h_d(\mathbf{E}, \text{Comp})}_{\text{exploitation/compensation}} + \underbrace{C_j^d(J_i)}_{\text{coupling}} + \underbrace{\eta \sum_j A_{ij} (J_j^d - J_i^d)}_{\text{diffusion}} - \underbrace{\kappa_d J_i^d}_{\text{decay}}$$

Internal:

$$g_D = \alpha_D A_i C_i, \quad g_P = \alpha_P A_i L_i, \quad g_R = \alpha_R S_i A_i, \quad g_C = \alpha_C A_i, \quad g_I = \alpha_I T_i L_i$$

Exploitation/compensation:

$$h_d = -\beta_d \sum_{i' \neq i} E_{i'i}^d + \beta'_d \sum_{i' \neq i} \text{Comp}_{i'i}^d$$

Corrective restorative:

$$\frac{dJ_j^c}{dt} += \alpha_c^{\text{rest}} \sum_i E_{ij}(t) \cdot \text{Comp}_{ij}(t)$$

4.3 Cross-Level Accountability (Dimension-Specific)

Let $m_i^{d,(l)}(t) = \min_{j \in \mathcal{C}(i)} J_j^{d,(l-1)}(t)$ denote the minimum value of dimension d across jurisdiction i 's children. Then:

$$A_i^{(l)} \leftarrow A_i^{(l)} + \lambda_c [\bar{m}_i^{(l)} - A_i^{(l)}] - \sum_d \lambda_p^d [J_{\min}^d - m_i^{d,(l)}]^+$$

where $\bar{m}_i^{(l)} = \frac{1}{|\mathcal{D}|} \sum_d m_i^{d,(l)}$ is the average of dimension-specific minimums.

Default values. $\lambda_c = 0.15$. Dimension-specific penalties:

Table 11. Default dimension-specific accountability penalties (λ_p^d)

Dimension	λ_p^d
Distributive	0.30

Dimension	λ_p^d
Procedural	0.30
Recognition	0.45
Corrective	0.20
Intergenerational	0.35

Elevated penalties for recognition and intergenerational justice reflect the view that these dimensions are systematically under-enforced. This is a normative choice, not a derivation.

Properties. No masking across dimensions; non-trivial coupling; consistency with C4. A child with high distributive justice and low recognition justice triggers the recognition penalty even when the aggregate is adequate.

4.4 Justice Suppression of Exploitation

$$\frac{dE_{ij}}{dt} = -\mu \cdot \min(J_i, J_j) \cdot E_{ij}$$

5. Welfare Function

5.1 Base Welfare

$$W(t) = \sum_{i,l} \phi_i^{(l)} F_i^{(l)}(t)$$

5.2 Dimension-Specific Justice-Enhanced Welfare

$$W^*(t) = W(t) + \sum_{i,l} \sum_d \gamma_d J_i^{d,(l)}(t) - \sum_{i,l} \sum_d \delta_d \max(0, J_{\min}^d - J_i^{d,(l)}(t))$$

The vectors $\gamma = (\gamma_D, \dots, \gamma_I)$ and $\delta = (\delta_D, \dots, \delta_I)$ encode value judgments about the relative moral importance of dimensions and their deficits. Setting $\gamma_d = \gamma \cdot w_d$ and $\delta_d = \delta \cdot w_d$ recovers the scalar form.

Default: $\gamma_d = 0.75 w_d$, $\delta_d = 2.0 w_d$.

5.3 Justice and Distributional Considerations

Following Rawls (1971), Sen (1999), Nussbaum (2011), and recent federalism scholarship (Sangiovanni, 2022; Miller, 2023), GSM-J recognizes that inequalities are permitted only if they benefit the least advantaged; fair equality of opportunity applies across jurisdictions; the difference principle extends to inter-jurisdictional relations; and compensation is a legitimate mechanism for inter-jurisdictional justice.

5.4 On the Multiplicity of Justice Roles

Justice enters in four roles:

Table 12. The four roles of justice in GSM-J

Role	Where	Function
State variable	$J_i^{(l)}$ in $\mathbf{x}$	Co-evolves with capabilities
Direct welfare contribution	$\sum_d \gamma_d J_i^d$	Constitutive of flourishing
Prioritarian penalty	$-\sum_d \delta_d [J_{\min}^d - J_i^d]^+$	Deficits reduce welfare asymmetrically
Mediator of exploitation	$-\mu \min(J_i, J_j) E_{ij}$	Suppresses exploitative dynamics

Each term operates on a distinct mechanism. Ablation scenarios S13–S15 decompose the total effect.

6. Simulation Protocol

6.1 Parameters

Table 13. Default parameter values for GSM-J simulations

PARAMETER	SYMBOL	ROLE	DEFAULT
JUSTICE WEIGHTS	w_d	Aggregate dimension weights	Simplex

JUSTICE FLOORS	$J_{\min}^d$	Per-dimension minima	0.10–0.30
AGGREGATE FLOOR	$J_{\min}^{\text{agg}}$	Threshold for C3	0.30
PRIORITARIAN PENALTY	δ_d	Dimension-specific penalties	$2.0 w_d$
WELFARE WEIGHT	γ_d	Dimension-specific contributions	$0.75 w_d$
SUBSIDIARITY	ϕ	Vertical non-regression	0.85
DIFFUSION	η	Horizontal diffusion	0.20
DECAY	κ_d	Per-dimension decay	0.05
INTERNAL COUPLING	α_d	Capability-to-justice coupling	0.20–0.35
RESTORATIVE	α_c^{rest}	Compensation-to-restoration	0.30
ACCOUNTABILITY COUPLING	λ_c	Parent–child convergence	0.15
ACCOUNTABILITY PENALTY	λ_p^d	Dimension-specific penalties	0.20–0.45
EXPLOITATION SUPPRESSION	μ	Justice suppression of exploitation	0.25
JUSTICE COUPLING	m_{de}	Signed coefficients	± 0.30 bound
INTERVENTION COST	κ^{int}	Per unit of correction	0.10
INTERVENTION HORIZON	τ^{int}	Steps to full implementation	5
INTERVENTION CHANNEL WEIGHTS	ω^k	Cost distribution	(0.50, 0.25, 0.15, 0.10)

6.2 Simulation Scenarios

Table 14. Simulation scenarios S0–S27

SCENARIO	DESCRIPTION
S0	GSM without justice cluster
S1	GSM-J default
S2–S5	Ablate C2, C4, diffusion, correction
S6–S7	Utilitarian and strict prioritarian baselines
S8	Weight sweep
S9–S10	Shock; historical injustice

S11–S12	No M_j ; recognition-only coupling
S13–S15	Isolate γ, δ, μ terms
S16–S17	Isolate accountability coupling, penalty
S18–S19	Soft-only, hard-only constraints
S20	Aggregate C4 mode
S21	Negative M_j disabled
S22	Dimension-specific welfare sweep
S23	High intervention cost
S24	Symmetric vs. asymmetric M_j
S25	Uniform vs. differentiated λ_p^d
S26–S27	Single-channel vs. multi-channel intervention cost

6.3 Metrics

Aggregate flourishing W^* ; flourishing variance; justice floor violations per dimension; exploitation persistence; convergence time; regime shifts; path dependence; intervention frequency and cumulative cost.

6.4 Constraint Enforcement Modes

Mode A — Constitutional projection via policy intervention (§6.5–6.6).

Mode B — Penalty dynamics.

Hybrid (default) — Per-constraint choice.

6.5 Policy Intervention: Cost, Delay, Feasibility

Projection Π is not an autonomous state update. It represents the collective effect of institutional decisions — legislation, constitutional amendment, court order, administrative reform, international agreement. Every projection carries:

1. A **designated authority** (C2: jurisdiction or parent; C3: exploiting jurisdiction; C4: parent).
2. An **implementation cost** κ^{int} per unit of correction.
3. A **finite horizon** τ^{int} (steps to full implementation).
4. A **feasibility condition** — projection is attempted only when the authority has sufficient capability.

6.6 Multi-Channel Capability Effects

Intervention cost $\kappa^{\text{int}} \cdot |\Delta|$ is distributed across four capability channels:

$$\omega = (\omega^C, \omega^S, \omega^T, \omega^A), \sum_k \omega^k = 1, \omega^k \geq 0$$

Capability update during intervention:

$$\Delta C_i = -\omega^C \frac{\kappa^{\text{int}} |\Delta|}{\tau^{\text{int}}}, \Delta S_i = -\omega^S \frac{\kappa^{\text{int}} |\Delta|}{\tau^{\text{int}}}, \text{ etc.}$$

Default weights:

Table 15. Multi-channel intervention cost weights (ω^k)

Channel	Weight	Rationale
ω^C	0.50	Administrative capacity is the primary constraint
ω^S	0.25	Interventions frequently produce short-term polarization
ω^T	0.15	Overturning prior commitments weakens continuity
ω^A	0.10	Override of local decision-making has modest accountability cost

Passive recovery. Once intervention horizons elapse, capability losses are gradually restored:

$$\frac{dC_i}{dt} = \rho_C (C_i^{\text{baseline}} - C_i)$$

with $\rho_k = 0.05$ default for each channel.

Consequences.

1. Interventions are not free even when technically feasible.
2. Recovery is slow relative to political cycles, so repeated interventions cumulatively depress capability.
3. **Intervention frequency ceiling.** Aggressive justice-improvement programs are self-limiting unless balanced with capability-restoration periods.

7. Hypotheses

Table 16. Hypotheses H1–H14

#	Hypothesis
H1	Stronger vertical accountability → stronger cross-level coordination
H2	Polycentric structures outperform monocentric on adaptive learning
H3	Well-calibrated information fidelity → better cross-level coordination
H4	Recursive learning at lower levels strengthens higher-level learning
H5	Bounded non-exploitation improves long-run flourishing
H6	Compensation reduces exploitation persistence
H7	Higher procedural justice → stronger adaptive learning
H8	Recognition justice moderates cohesion–exploitation relationship
H9	Intergenerational justice negatively correlated with short-term welfare maximization
H10	Justice diffusion reduces cross-level exploitation
H11	Justice floors reduce variance of flourishing outcomes
H12	Dimension-specific accountability penalties reduce recognition deficits more effectively than aggregate penalties
H13	Multi-channel intervention costs produce a frequency ceiling on justice-improving interventions
H14	Asymmetric M_j estimation improves fit when directional hypotheses hold

8. Python Implementation: `gsm-governance`

Open-source library, six modules: `gsm.core`, `gsm.justice`, `gsm.enforcement`, `gsm.welfare`, `gsm.simulation`, `gsm.reporting`. Core class signatures:

```

# python
@dataclass
class NormativeParams:
    justice_weights: Dict[str, float]
    justice_floors: Dict[str, float]
    aggregate_floor: float
    gamma_dim: Dict[str, float]
    delta_dim: Dict[str, float]
    lambda_p_dim: Dict[str, float]
    phi: float

@dataclass
class StructuralParams:
    eta, kappa, alpha, alpha_corrective_restorative: ...
    beta_erosion, beta_compensation: float
    lam_c, mu: float
    intervention_cost: float
    intervention_horizon: int
    intervention_weights: Dict[str, float]
    recovery: Dict[str, float]

class JusticeInteractionMatrix:
    m: Dict[Tuple[str, str], float]
    m_max: float
    regime: MJRegime
    def coupling(self, J: Dict[str, float], d: str) -> float: ...
    def symmetrize(self) -> None: ...

class JusticeCapability:
    d: Dict[str, float]
    def aggregate(self, weights) -> float: ...
    def satisfies_floors(self, floors) -> bool: ...
    def step(self, ...) -> None: ...

class PolicyInterventionEnforcer:
    def check_and_schedule(self, jurisdictions, exploitation) -> None: ...
    def apply_step(self, jurisdictions, exploitation) -> None: ...
    def apply_recovery(self, jurisdictions) -> None: ...
    def enforce(self, jurisdictions, exploitation) -> None: ...

```

```
def welfare_dimension_specific(base_welfare, jurisdictions, n) -> float: ...  
def update_accountability_dimension_specific(parent, jurisdictions, n, s) -> None: ...
```

Full source, tests, and worked examples are in the repository.

9. Stylized Demonstration Scenario

A six-level governance system with 98 stylized municipalities, 5 regions, one national level, and one EU-level jurisdiction. Initial capability profiles reflect a plausible high-capability system. **No claim is made that these values correspond to actual data.**

9.1 Qualitative Results

Under default parameters, the model converges to high distributive and procedural justice, moderate recognition, moderate corrective, and moderate intergenerational justice. Aggregate welfare rises modestly when justice is added (S1 vs. S0).

Four findings distinguish the current version:

- 1. Dimension-specific accountability (S25).** Under uniform λ_p^d , recognition justice converges more slowly. Differentiated penalties produce measurably faster recognition improvement because aggregate penalties would permit masking.
- 2. Multi-channel intervention costs (S26 vs. S27).** Under single-channel costs, aggressive intervention can be sustained at high frequency. Under multi-channel costs, cohesion and continuity deplete materially, producing an endogenous frequency ceiling after roughly 15–20 interventions in quick succession.
- 3. Symmetric vs. asymmetric M_j (S24).** When the data-generating process is asymmetric, forcing symmetry biases welfare trajectories by up to 22% in the stylized run.
- 4. Negative M_j entries (S21).** Disabling tension entries produces higher aggregate justice but *lower* long-run welfare, because tension entries prevent over-investment in single dimensions.

9.2 Justice–Flourishing Trade-offs

Under strict prioritarianism (S7), present welfare falls slightly but long-run welfare rises substantially. Under utilitarianism (S6), present welfare is maximized but intergenerational justice falls below the floor.

9.3 From Stylized Demonstration to Empirical Calibration

Full empirical calibration would require indicator data per dimension per level, estimation of M_j under §3.3.2, estimation of exploitation from trade/fiscal/regulatory/ecological data, and validation against held-out data. None is performed here.

10. Discussion

10.1 What the Model Shows

Justice is not reducible to non-exploitation. A jurisdiction may satisfy bounded non-exploitation while violating the justice floor, exploiting a vulnerable neighbor, or eroding child justice.

Convergence conditions. Justice and flourishing converge when justice is treated as a capability, exploitation is justice-decomposed, corrective dynamics are active, subsidiarity is enforced dimension-wise, and cross-level accountability is dimension-specific.

Where they conflict. Intergenerational justice constrains present optimization; corrective justice reduces present consumption; recognition justice may reduce short-run cohesion; multi-channel intervention costs imply that aggressive justice improvement is self-limiting.

10.2 Normative Commitments

Table 17. Normative commitments exposed by GSM-J parameters

Parameter	What it encodes
w_d	Dimension importance in aggregate
$J_{\min}^d, J_{\min}^{\text{agg}}$	Moral minimum per dimension
γ_d, δ_d	Dimension-specific welfare weights
λ_p^d	Dimension-specific accountability penalties
ϕ	Tolerated vertical regression
τ_j	Tolerance for exploitation
$M_j, m_{\max}, \sigma_{de}$	Interaction matrix, bounds, sign priors

Parameter	What it encodes
$\omega^k, \kappa^{\text{int}}, \tau^{\text{int}}$	Intervention cost structure

10.3 Policy Implications

1. Institutional design should target justice as a capability.
2. Justice floors should be dimension-specific.
3. Vulnerability-sensitive non-exploitation is stronger than bounded non-exploitation.
4. Corrective mechanisms must build capability.
5. Subsidiarity should be enforced dimension-wise.
6. Constraint enforcement modes should be explicit.
7. Justice dimensions interact via both synergy and tension.
8. Projection is costly intervention, not instantaneous adjustment.
9. Accountability penalties should be dimension-specific.
10. Intervention costs span multiple capabilities; aggressive justice improvement is self-limiting.

10.4 Positioning within Computational Normative Governance

GSM-J sits at the intersection of computational social science (Lazer et al., 2020; Watts, 2021), computational public policy (Athey, 2018; Kleinberg et al., 2018; Mullainathan & Obermeyer, 2022), algorithmic governance (Yeung, 2018; Danaher, 2019; Citron, 2022), and digital democracy (Landemore, 2020; Fishkin, 2021). Its contribution is a formal substrate in which normative commitments can be formalized, parameterized, and examined for systemic consequences.

11. Limitations

1. **Normative non-derivability.** GSM-J formalizes justice; it does not derive its commitments.
2. **Weight sensitivity.** Results depend on many parameters; the scenario protocol sweeps them.
3. **Measurement validity.** Recognition and intergenerational indicators are less developed than distributive ones.
4. **Levels of aggregation.** Household-level justice is not fully represented.
5. **Cultural pluralism.** Justice dimensions may vary across contexts.
6. **Dynamics calibration.** Functional forms are stylized; empirical estimation of M_J is deferred.
7. **Tragic conflict.** Trade-offs are represented but not resolved.
8. **Path dependence.** Different initial profiles may yield different equilibria.
9. **Multiplicity of justice roles.** Defended in §5.4, but boundaries are philosophically contested.
10. **Constraint enforcement.** The hybrid default is a design choice.
11. **Projection interpretation.** The mapping from real institutional processes to $\kappa^{\text{int}}, \tau^{\text{int}}, \omega^k$ is not fully specified.

12. **M_J identification.** In datasets lacking exogenous variation or sign restrictions, identification may be weak.
13. **Multi-channel intervention weights.** Default ω^k are theoretically motivated but not empirically validated.
14. **Stylized demonstration.** The demonstration is illustrative, not empirical.

12. Conclusion

GSM-J extends the GSM in eight ways: justice as a state variable; a five-dimensional decomposition; a signed interaction matrix with disciplined estimation; three additional constraints including vulnerability-sensitive non-exploitation; dimension-specific vertical alignment; dimension-specific cross-level accountability; a dimension-specific prioritarian welfare function; and constraint projection reframed as multi-channel policy intervention with designated authority, cost, finite horizon, and passive recovery.

The result is a conditional normative systems model: descriptive about mechanisms, explicit about values, silent about which values are correct. Its scientific contribution is the formal structure. Its ethical contribution is making the specification unavoidable. Its practical contribution is a simulation architecture in which justice and flourishing can be jointly examined, contested, and compared.

The architecture is now complete. Further development should focus on empirical calibration, not theoretical expansion.

Appendix A — Notation

Table 18. Notation used throughout the paper

Symbol	Meaning
A, C, S, T, L	Accountability, competence, cohesion, continuity, learning
J, J^d	Justice capability; dimension $d \in \{D, P, R, C, I\}$
w_d	Justice dimension weights
$J_{\min}^d, J_{\min}^{agg}$	Justice floors
$M_J, m_{de}, m_{\max}, \sigma_{de}$	Interaction matrix, coefficients, bound, sign priors
γ_d, δ_d	Dimension-specific welfare weights
E_{ij}, E_{ij}^d	Exploitation index
$Comp_{ij}$	Compensation vector
τ_j, ϕ	Tolerance threshold; subsidiarity
λ_c, λ_p^d	Accountability coupling; dimension-specific penalties
$m_i^{d,(l)}$	Dimension-specific minimum child justice
$\kappa^{int}, \tau^{int}, \omega^k, \rho_k$	Intervention cost, horizon, channel weights, recovery
W, W^*	Base welfare; justice-enhanced welfare

Appendix B — Reporting Checklist

- Full vector w_d with justification
- Full vector $J_{\min}^d$ with justification
- γ_d, δ_d with justification
- λ_p^d with justification
- ϕ, τ_j
- Signed M_j with sources and estimation regime
- $m_{\max}$ and sign priors
- Recognition justice measurement protocol
- Constraint enforcement mode and C4 mode
- $\kappa^{\text{int}}, \tau^{\text{int}}, \omega^k, \rho_k$
- Sensitivity analysis over all of the above

Appendix C — Measurement Protocol for Recognition Justice

C.1 Non-domination R^{dom} . Republican Freedom Index (12 items), 5-point Likert, CFA aggregation.

C.2 Non-exclusion R^{exc} . Representation ratios, access indicators, exclusion events. Composite adjusted for group size and historical exclusion burden.

C.3 Non-assimilation R^{assim} . Language retention (0.4), cultural practice freedom (0.3), assimilation pressure inverse (0.2), recognition events (0.1).

C.4 Reliability. Cultural specificity, self-report bias, aggregation loss, conceptual contestation. Recommend triangulation, cross-cultural piloting, longitudinal and participatory validation.

Appendix D — Identification Protocol for M_j

D.1 Parameter space. 20 off-diagonal coefficients (asymmetric) or 10 (symmetric), bounded to $[-, 0.30 + 0.30]$.

D.2 Sign priors. Each coefficient receives $\sigma_{de} \in \{+, -, \pm, 0\}$.

D.3 Observational model. Indicators observed with Gaussian noise.

D.4 Estimation. Hierarchical Bayesian dynamic factor model with particle filter or EKF inference; group-level m_{de}^{group} with jurisdiction-specific deviations.

D.5 Identification. Exogenous shocks per jurisdiction; sufficient temporal variation; sign restrictions on a subset of coefficients.

D.6 Reporting. Posterior means, 90% credible intervals, posterior predictive checks, sensitivity to $m_{\max}$, sign prior strength, and symmetric vs. asymmetric regime.

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